INTERNSHIP *PROJECT* REPORT

BhashaDaan Data Analysis

BHASHINI

Submitted By

SNEHA SINGH

DECLARATION

I, Sneha Singh, hereby declare that the internship report titled "Internship Project Report", submitted to the Office of The Chief Executive Officer, Digital India BHASHINI Division (DIBD), is a record of original work done by me during my two-month internship at the Digital India BHASHINI Division (DIBD), MeitY, Government of India, New Delhi, under the guidance of the DIBD technical mentorship team.

I further declare that the information, research findings, system designs, and technical contributions presented in this report – including BhashaDaan Data Analysis - have been carried out by me, and wherever external references, frameworks, or prior works have been used, they have been duly cited and acknowledged.

This report has not been submitted partially or wholly to any other university, institution, or organization for the award of any degree, diploma, or certification.

SNEHA SINGH
DATE :01/06/26

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Lastly, I would like to thank all my colleagues, peers, and well-wishers whose support and encouragement made this internship experience enriching and memorable.

Yours faithfully,
Sneha Singh
Intern
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BHASHADAAN FEEDBACK ANALYSIS REPORT

1. INTRODUCTION

BHASHADAAN is a crowdsourcing platform developed under the Digital India BHASHINI initiative to support the collection, validation, and enhancement of multilingual language resources. The platform enables users to contribute language data and interact with various language technologies such as Translation, Automatic Speech Recognition (ASR), Text-to-Speech (TTS), and Language Detection.

As the platform continues to grow, it becomes important to evaluate system performance, user engagement, and service quality. Feedback analysis provides valuable insights into the effectiveness of deployed language models, the popularity of different language tasks, language-wise usage patterns, and the alignment between automated system ratings and actual user experiences.

This report presents a comprehensive analysis of feedback data collected from the BHASHADAAN platform. The study focuses on identifying usage trends, understanding language distribution across different tasks, evaluating model adoption, and assessing the consistency between pipeline-generated quality ratings and user feedback ratings.

The findings of this analysis can support future improvements in model performance, user satisfaction, and platform-wide quality assurance mechanisms.

2. OBJECTIVES OF THE ANALYSIS

The primary objective of this study is to analyze feedback data generated through the BHASHADAAN platform and derive meaningful insights regarding user engagement, service utilization, language diversity, and quality assessment mechanisms.

The specific objectives of this analysis are:

- To identify the most frequently used language processing models deployed on the platform.
- To analyze the distribution of different language technology tasks, including Translation, Automatic Speech Recognition (ASR), Text-to-Speech (TTS), and Language Detection.
- To study language-wise participation patterns across various tasks and identify dominant source languages.

- To evaluate task-specific language preferences and understand how different languages contribute to platform activity.
- To assess the effectiveness of pipeline-generated quality ratings by comparing them with actual user feedback ratings.
- To identify discrepancies between automated evaluation mechanisms and human user experiences.
- To examine overall platform usage trends and determine which services contribute most significantly to user activity.
- To provide recommendations for improving service quality, user satisfaction, and language coverage.
- To identify opportunities for future enhancement of language technologies within the BHASHINI ecosystem.
- To generate actionable insights that support data-driven decision-making and continuous platform improvement.

The analysis aims to bridge the gap between system-level performance metrics and actual user experiences, thereby contributing toward the development of more reliable, accurate, and user-centric multilingual AI services.

3. OVERVIEW OF BHASHADAAN PLATFORM

3.1 About BHASHINI

BHASHINI (BHASHa INterface for India) is a flagship initiative of the Government of India under the Ministry of Electronics and Information Technology (MeitY). The mission of BHASHINI is to enable seamless digital access in Indian languages through Artificial Intelligence, Natural Language Processing (NLP), Speech Technologies, and multilingual computing solutions.

The initiative seeks to bridge linguistic barriers by providing language technologies that can support communication, education, governance, and digital inclusion across India's diverse linguistic landscape.

3.2 About BHASHADAAN

BHASHADAAN is a community-driven crowdsourcing platform developed under the BHASHINI ecosystem. The platform enables users to contribute, validate, and improve multilingual language datasets that are essential for building and enhancing language AI systems.

The platform relies on active participation from users who contribute speech, text, and validation data in multiple Indian languages. These contributions help improve the quality and accuracy of AI models deployed across various language technology services.

3.3 Core Services Supported by BHASHADAAN

The platform supports several language technology services including:

Machine Translation

Machine Translation enables automatic translation of text between different languages. This service facilitates multilingual communication and increases accessibility of digital content.

Automatic Speech Recognition (ASR)

ASR systems convert spoken language into text. This technology is particularly useful for voice-based applications, transcription services, and accessibility solutions.

Text-to-Speech (TTS)

TTS systems generate natural-sounding speech from textual input. These systems support accessibility applications, educational tools, and voice-enabled digital services.

Language Detection

Language Detection systems automatically identify the language of a given text input, enabling efficient routing and processing of multilingual content.

3.4 Importance of Feedback Analysis

As language technologies continue to evolve, user feedback plays a crucial role in evaluating service quality and identifying areas for improvement. Feedback analysis helps measure user satisfaction, assess model performance, identify language-specific challenges, and validate automated quality assessment mechanisms.

The insights obtained through feedback analysis contribute directly to improving system accuracy, enhancing user experience, and strengthening multilingual AI infrastructure.

4. DATASET DESCRIPTION

4.1 Dataset Overview

The dataset analyzed in this report consists of feedback records generated through user interactions with BHASHADAAN services. These records capture information related to language processing requests, service utilization patterns, language preferences, and quality assessments.

The dataset serves as an important source of information for understanding how users engage with various language technologies offered through the platform.

4.2 Data Sources

The data was collected from platform-generated logs and feedback records associated with language technology services. These records provide information regarding:

- Service requests
- Task categories
- Language metadata
- Model identifiers
- User feedback ratings
- Pipeline-generated quality ratings

The dataset represents actual platform activity and provides a realistic view of user behavior and system performance.

4.3 Key Attributes Used in Analysis

Several important features were utilized during the analysis process.

Service ID

Represents the language model or service responsible for processing a user request.

Task Type

Specifies the language technology service being used. Examples include:

- Translation
- Automatic Speech Recognition (ASR)
- Text-to-Speech (TTS)
- Language Detection

Source Language

Identifies the language provided as input by the user.

Pipeline Rating

Represents the automated quality score generated internally by the platform.

Feedback Rating

Represents the quality rating assigned by users based on their experience with the generated output.

4.4 Data Preparation

Before performing analysis, several preprocessing steps were applied to ensure data quality and consistency.

The preprocessing workflow included:

- Removal of duplicate records
- Handling missing values
- Standardization of language codes
- Standardization of task labels
- Verification of rating values
- Validation of service identifiers

These preprocessing steps improved the reliability of subsequent analyses and reduced the possibility of misleading conclusions.

4.5 Scope of Analysis

The analysis focuses on:

- Service utilization patterns
- Task distribution
- Language-wise activity
- Model popularity
- Rating consistency
- User experience indicators

The study does not attempt to evaluate the internal architecture of language models but instead focuses on observable platform behavior and user interactions.

5. METHODOLOGY

5.1 Analytical Approach

A structured data analytics approach was adopted to examine user feedback and service utilization patterns within the BHASHADAAN platform.

The methodology combines descriptive statistics, exploratory data analysis, and quality assessment techniques to generate meaningful insights from the dataset.

The overall workflow consisted of:

Dataset Collection → Data Cleaning → Exploratory Analysis → Statistical Evaluation → Visualization → Insight Generation

5.2 Data Cleaning and Preprocessing

Raw datasets often contain inconsistencies that can affect analytical results. Therefore, a preprocessing stage was conducted prior to analysis.

The following activities were performed:

- Identification of missing values
- Removal of duplicate records
- Validation of task categories
- Standardization of language identifiers
- Verification of rating values
- Consistency checks across service records

This step ensured that the dataset was suitable for reliable analysis.

5.3 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to identify trends, patterns, and anomalies within the dataset.

The EDA process focused on:

- Model utilization frequency
- Distribution of language tasks
- Language participation patterns
- Rating distributions

• Task-wise service consumption

The objective was to understand the overall structure of the data before performing deeper investigations.

5.4 Statistical Analysis

Descriptive statistical methods were used to summarize platform activity and evaluate service performance.

Key metrics analyzed include:

• Service request counts

• Language frequencies

• Task frequencies

• Rating agreement rates

• Distribution percentages

These metrics provide quantitative insights into user behavior and platform utilization.

5.5 Visualization Techniques

Visual representations were generated to improve interpretability of analytical findings.

The following visualization techniques were utilized:

• Bar Charts

• Distribution Plots

• Frequency Graphs

• Comparative Analysis Charts

• Task-wise Language Distribution Charts

These visualizations help highlight major trends and simplify communication of analytical results.

5.6 Rating Consistency Analysis

One of the most important components of this study involved comparing automated pipeline ratings with actual user feedback ratings.

The comparison aimed to determine:

• Whether automated quality scores accurately reflect user satisfaction.

• The level of agreement between machine-generated evaluations and human assessments.

• Potential weaknesses in the existing quality assessment mechanism.

This analysis serves as a critical indicator of platform reliability and user trust.

5.7 Tools and Technologies Used

The analysis was conducted using Python-based data analytics tools.

The primary technologies used include:

• Python

• Pandas

• NumPy

• Matplotlib

• Jupyter Notebook

These tools enabled efficient data processing, visualization, and statistical analysis.

6. ANALYSIS AND FINDINGS

6.1 BHASHADAAN USER DATA ANALYSIS REPORT (17-01-26 to 17-04-26)

1. SUMMARY

- **User Registration Data**: A total of **91 records** were captured in the User Registration spreadsheet. This reflects the number of individuals who signed up to engage with the platform during the period.
- **Contribution Data**: The Contribution Counts spreadsheet recorded **571 entries**, representing the volume of user activity and participation across the platform.
- **Insights**: The data highlights both the growth in user registrations and the level of contributions made. This dual perspective provides valuable understanding of how users are interacting with Bhashadaan.
-

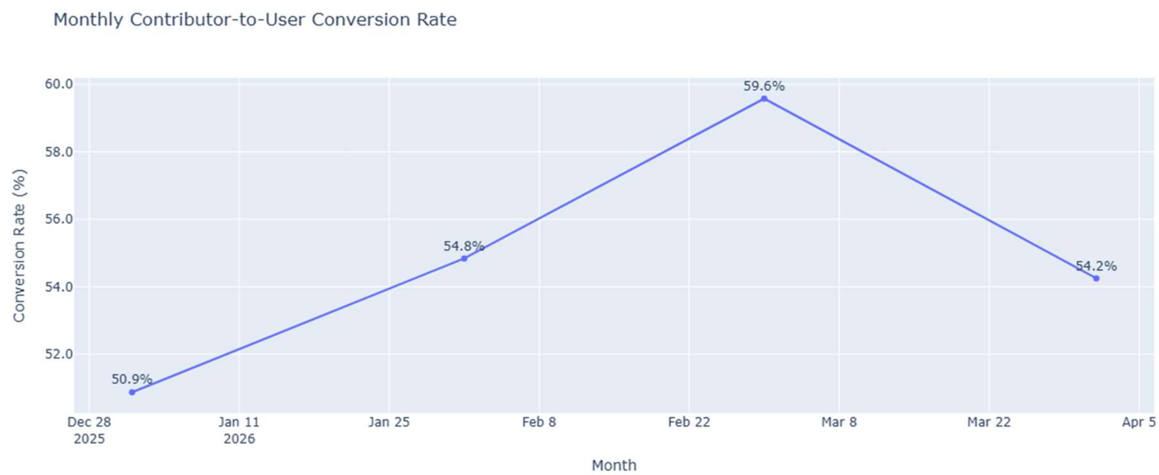
2. INSIGHT REPORT

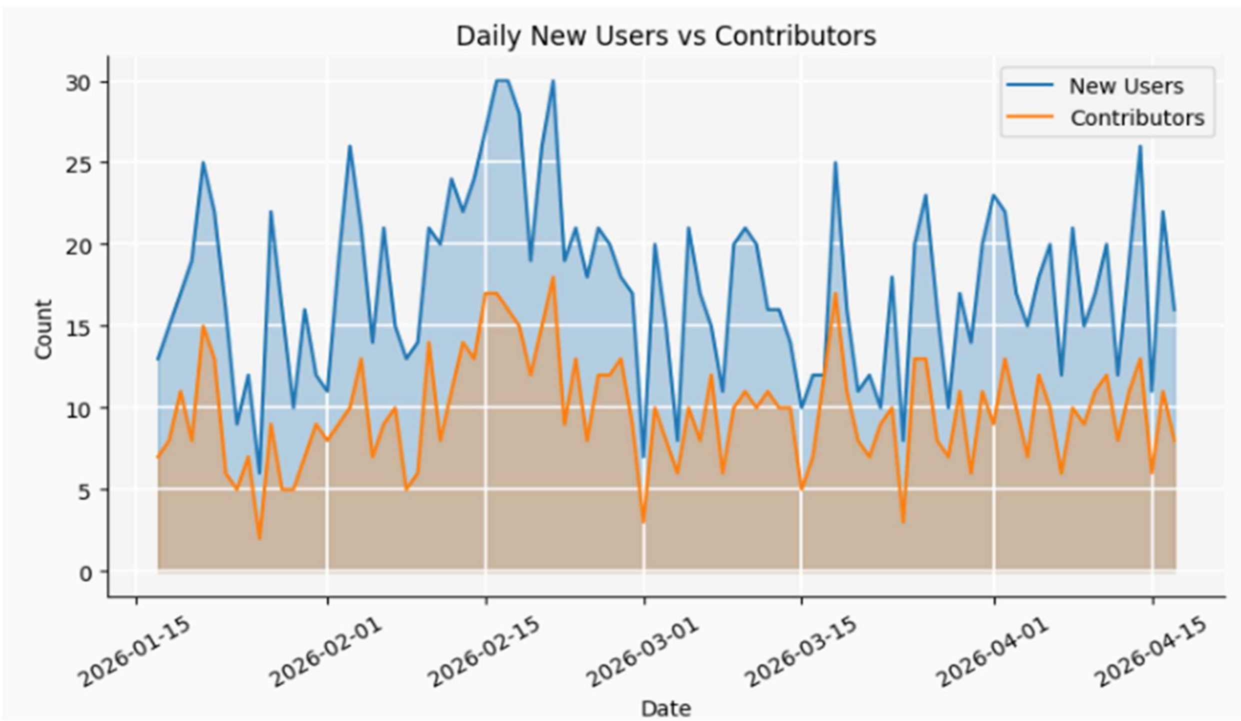
2.1 User to Contributor:

Best Converting Month: March 2026 Conversion Rate: 59.6% New Users: 475 Contributors: 283

Lowest Converting Month: January 2026 Conversion Rate: 50.9% New Users: 230 Contributors: 117

The upward trend from January to March indicates improved user engagement and effectiveness of onboarding strategies, although a slight decline is observed post-March.





Most new users registered : 2026-02-16 (30)

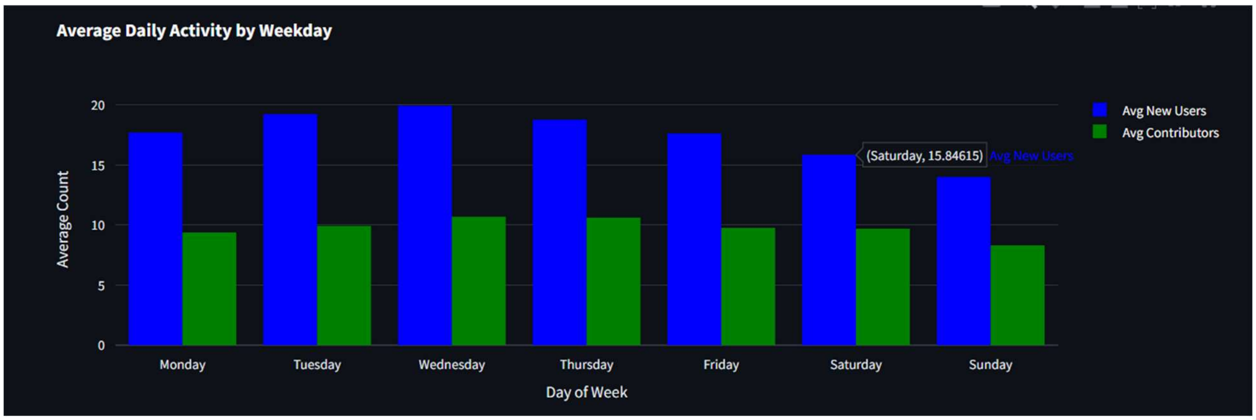
Most contributors recorded : 2026-02-21 (18)

2.2Contributor analysis daily and weekly basis

Top activity days of users recorded

	Activity_date	New_users	Contributors	User_to_Contributor(%)
0	2026-02-21	30	18	60.000000
1	2026-02-16	30	17	56.666667
2	2026-02-15	27	17	62.962963
3	2026-03-18	25	17	68.000000
4	2026-02-17	30	16	53.333333

Mostly the conversion rates are higher than the average. Notably, **18 March 2026** achieved the highest conversion rate. In general February has most activity recorded.



The average daily activity across weekdays shows a clear behavioral trend:

- **Mid-week (Tuesday–Thursday)** records the highest user and contributor activity.
- **Wednesday** shows peak engagement in terms of both new users and contributors.
- **Weekends (Saturday–Sunday)** experience a noticeable drop in activity.

2.3 Monthly Contribution Analysis :

Month_Year	Activity_date	Language	Contributions
April 2026	2026-04-10	English	112
February 2026	2026-02-02	Odia	371
January 2026	2026-01-20	English	108
March 2026	2026-03-27	Telugu	380

These are the dates in each month on which most activity was reported

- **March 2026 recorded the highest single-day contribution (380)**, closely followed by February. Meanwhile overall February has recorded the most contributions
- A significant surge in contributions is observed in **regional languages such as Odia and Telugu**, indicating strong participation beyond dominant languages.

	Month_Year	Unique_Languages
0	April 2026	23
1	February 2026	21
2	January 2026	20
3	March 2026	21

The number of unique languages contributed each month reflects the diversity of user participation across the platform. In terms of language diversity April has most diverse contributions.

2.4 Language Diversity

Over the last four months, the majority of contributions were concentrated in:

- **English** (30%)
- **Hindi** (12%)
- **Kashmiri** (12%)

Language with most contributions: English 3524
Language with most validations: English 2930

	Language	Contributions	Validations	Total	Pct_of_Total
5	English	3524	2930	6454	30.648685
7	Hindi	1782	903	2685	12.750499
23	Telugu	1108	747	1855	8.809004
9	Kashmiri	1055	1555	2610	12.394339
0	Assamese	612	25	637	3.024979
6	Gujarati	563	1439	2002	9.507076
17	Odia	481	278	759	3.604331
14	Marathi	367	232	599	2.844525
20	Santali	340	260	600	2.849273
12	Malayalam	307	77	384	1.823535
22	Tamil	261	123	384	1.823535
1	Bengali	150	125	275	1.305917
8	Kannada	120	118	238	1.130212
18	Punjabi	94	55	149	0.707570
19	Sanskrit	86	70	156	0.740811
4	Dogri	67	298	365	1.733308
24	Urdu	64	12	76	0.360908
21	Sindhi	55	2	57	0.270681
3	Bodo	38	538	576	2.735302
10	Konkani	20	8	28	0.132966
16	Nepali	16	14	30	0.142464
13	Manipuri	4	1	5	0.023744
15	Mavchi	1	0	1	0.004749
11	Maithili	0	128	128	0.607845
2	Bhili	0	5	5	0.023744

2.5 Contributions vs Validations

- **Consistent Gap Across Months:**

  In every month, **contributions exceed validations**, indicating that not all submitted data is immediately validated. This reflects the natural lag in the validation pipeline.

- **Largest Gap in High Activity Period (February):**

  February shows the **widest gap** between contributions (4831) and validations (4243).

  → This suggests that during peak activity, the **validation system struggles to keep pace** with incoming data.

- **Gap Reduces Over Time:**

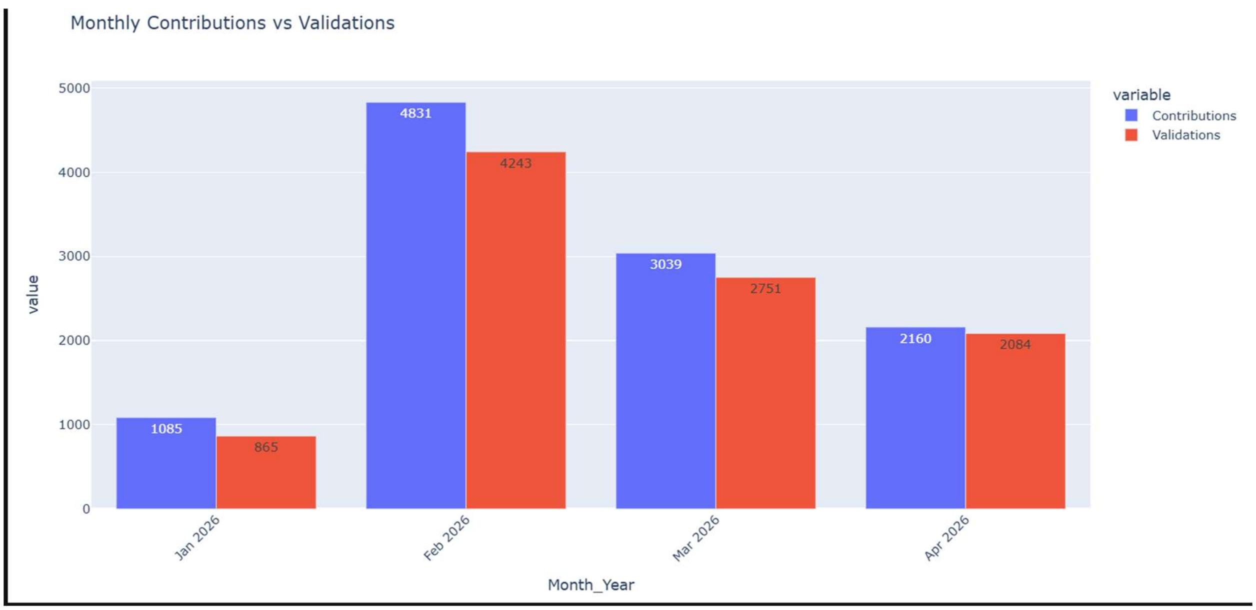
  From February → March → April, the difference between contributions and validations steadily decreases.

  → This indicates **improved validation efficiency** or reduced pressure on the system.

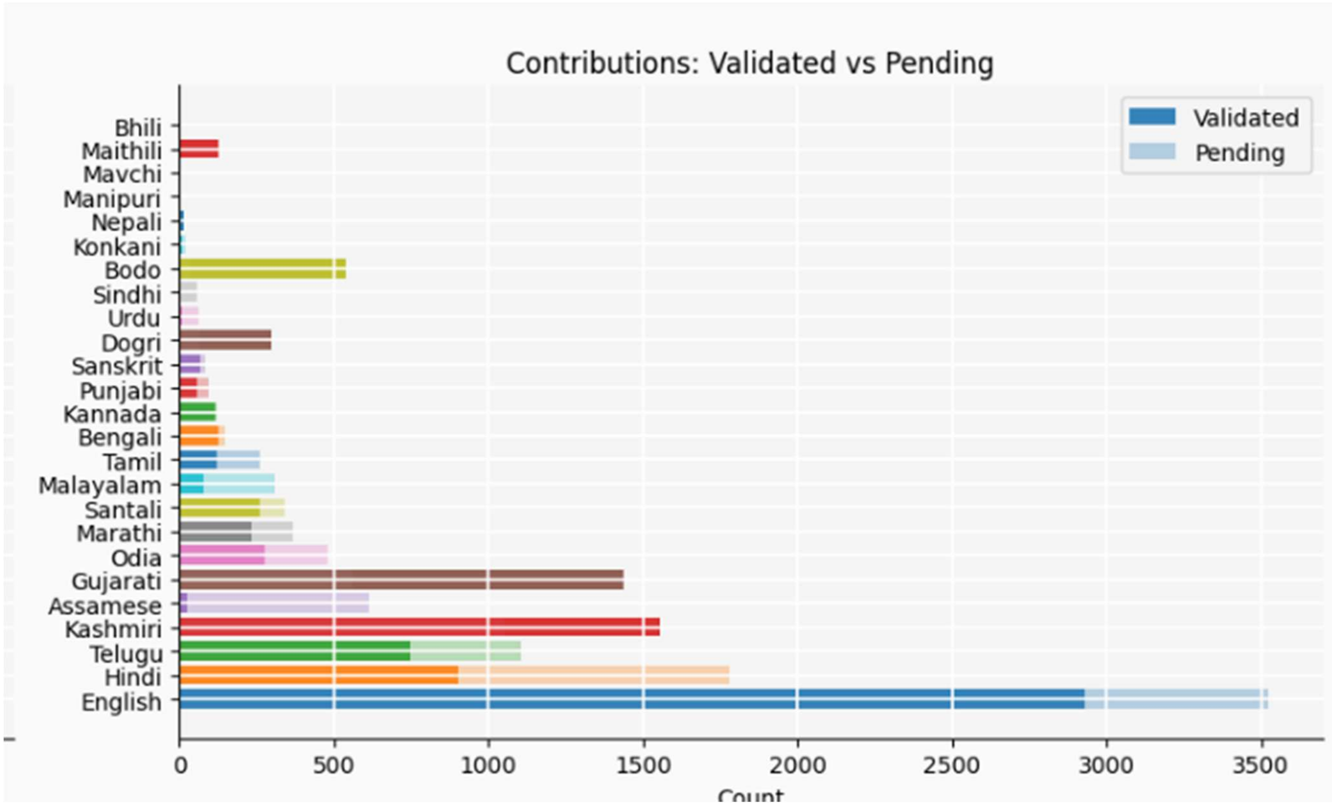
- **April Shows Near Balance:**

  In April, validations (2084) are very close to contributions (2160).

  → This suggests a **well-balanced system** where most contributions are being processed and validated promptly.



2.6 Language-wise Validation



Some languages have a large gap where contributions are much higher than validations, meaning a lot of data is still unverified.

1. High Backlog Languages (Low Validation Coverage)

Some languages have a **large gap where contributions are much higher than validations**, meaning a lot of data is still unverified:

- **Hindi:** 1782 contributions vs 903 validations (~49% pending)
- **English:** 3524 vs 2930 (~17% pending, but large absolute backlog)
- **Kashmiri:** 1055 vs 1555 *(interesting case, see below)*
- **Telugu:** 1108 vs 747 (~32% pending)
- **Odia:** 481 vs 278 (~42% pending)
- **Malayalam:** ~75% still pending

2. Over-Validated Languages (Validations > Contributions)

Some languages show **more validations than contributions**, resulting in negative "remaining" values:

- **Bodo, Dogri, Gujarati, Kashmiri, Maithili**
- **Bodo & Dogri** show extreme negative percentages → likely backlog clearance or data anomaly
- **Maithili** has 0 contributions but 128 validations.

	Language	Contributions	Validations	Remaining	Pct_Remaining
0	Assamese	612	25	587	95.915033
1	Bengali	150	125	25	16.666667
2	Bhili	0	5	-5	-inf
3	Bodo	38	538	-500	-1315.789474
4	Dogri	67	298	-231	-344.776119
5	English	3524	2930	594	16.855846
6	Gujarati	563	1439	-876	-155.595027
7	Hindi	1782	903	879	49.326599
8	Kannada	120	118	2	1.666667
9	Kashmiri	1055	1555	-500	-47.393365
10	Konkani	20	8	12	60.000000
11	Maithili	0	128	-128	-inf
12	Malayalam	307	77	230	74.918567
13	Manipuri	4	1	3	75.000000
14	Marathi	367	232	135	36.784741
15	Mavchi	1	0	1	100.000000
16	Nepali	16	14	2	12.500000
17	Odia	481	278	203	42.203742
18	Punjabi	94	55	39	41.489362
19	Sanskrit	86	70	16	18.604651
20	Santali	340	260	80	23.529412
21	Sindhi	55	2	53	96.363636
22	Tamil	261	123	138	52.873563
23	Telugu	1108	747	361	32.581227

Fig : Highlighted data shows ambiguity in validation and contribution data.

RECOMMENDATIONS

- **Track Active Users & Churn Rate:**
  Introduce a “current active users” metric to monitor retention and identify drop-off points in user engagement.
- **Capture User Location Data:**
  Adding location information will help map **region-wise language contributions** and target specific linguistic communities more effectively.
- **Enhance Language Diversification:**
  Focus on increasing contributions in underrepresented regional languages to better align with the platform’s inclusivity goals.

CONCLUSION

- **February 2026** recorded the highest overall interactions, marking the peak of user engagement on the platform.
- A **declining trend in contributions** is observed after February, indicating a drop in sustained user activity.
- **March 27, 2026** recorded the **highest single-day contribution**, highlighting strong but short-term activity spikes.
- **April 2026**, despite lower overall contributions, showed the **highest language diversity**, indicating broader participation across different linguistic groups.
- The majority of contributions were concentrated in **English, Hindi, and Kashmiri**, making them the dominant languages on the platform.
- Mostly users contribute in English and hindi language which adds little value in our mission.
- A **significant gap between contributions and validations** is observed, especially during high-activity periods like February. This suggests that the **validation system struggles to keep pace with incoming data**, leading to backlog accumulation.
- Found ambiguity in data where in some cases language validation dataset exceeds number of contribution dataset.

- Overall, the platform demonstrates **strong engagement with periodic peaks**, along with **growing multilingual participation**, indicating potential for both improved retention and expanded language outreach.

6.2 BHASHADAAN FEEDBACK ANALYSIS REPORT

1.TOP MODELS USED

A total of six core models were tracked based on service requests. The model ai4bharat/indictrans-v2-all-gpu--t4 processed the vast majority of the volume.

service_id

ai4bharat/indictrans-v2-all-gpu--t4	1496
bhashini/iiith/nmt-all	403
ai4bharat/indictrans-v2-all-gpu--t4/btp	125
Bhashini/IIITH/Trans/V2	120
Bhashini/IIITH/Trans/V1	93
Bhashini/IITM/TTS	58

2. TASK DISTRIBUTION ANALYSIS

The platform supports four core language processing tasks: Translation, Automated Speech Recognition (ASR), Text-to-Speech (TTS), and Text Language Detection. **Translation** constitutes the overwhelming majority of all platform activity

```
task_type
translation           2498
asr                    116
tts                    109
txt-lang-detection       2
Name: count, dtype: int64
```

3. LANGUAGE BREAKDOWN BY TASK TYPE

```
================================================================================
TASK: TTS
================================================================================
source_language
te     70
mai    18
en     12
as      5
hi      3
ml      1
Name: count, dtype: int64

================================================================================
TASK: TXT-LANG-DETECTION
================================================================================
```

```
================================================================================
TASK: TRANSLATION
================================================================================
source_language
en       2380
ml         71
hi         19
bn          8
ks          8
mr          4
ur          2
as          2
te          2
বাংলা       1
kn          1
Name: count, dtype: int64

================================================================================
TASK: ASR
================================================================================
source_language
en    38
ks    38
te    31
ml     5
bn     4
Name: count, dtype: int64
```

I. Task: Translation
English (en) is overwhelmingly the primary source language for translation workloads.

II. Task: ASR (Automatic Speech Recognition)

Speech recognition workloads show a more distributed language mix between English and Kashmiri (ks).

- **en**: 38
- **ks**: 38

- **te**: 31
- **hi**: 5
- **bn**: 4

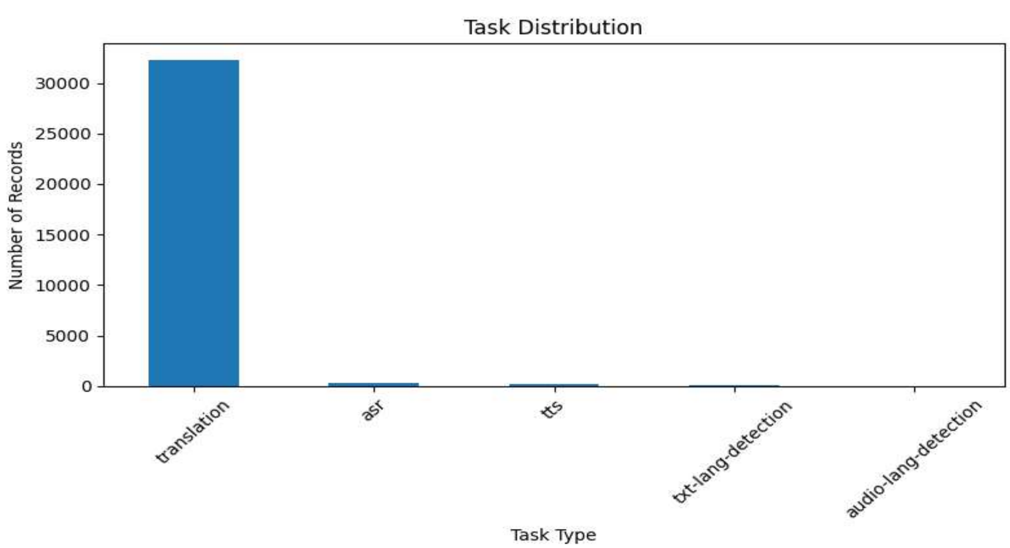
III. Task: TTS (Text-to-Speech)

Telugu (te) leads the text-to-speech generation requests, followed by Maithili (mai).

- **te**: 70
- **mai**: 18
- **en**: 12
- **as**: 5
- **hi**: 3
- **ml**: 1

IV. Task: TXT-LANG-DETECTION

- *Note: This task recorded a negligible volume (2 requests total) during this logging period, indicating it is either a secondary feature or rarely invoked by end-users.*



4.Rating Alignment & Pipeline Consistency

*A critical metric for evaluating the platform's automated quality assessment is comparing the system's **Pipeline Rating** against actual **User Feedback Ratings**.*

***Key Finding:** Out of 1,632 total compared rows, the pipeline rating and feedback ratings matched **only 20 times**. This extremely low consistency rate (1.23%) suggests a severe mismatch between how the* *internal system calculates pipeline quality and how human users experience the output.*

```
Total Compared Rows: 1632
Matches: 20
```

5. Task-Wise Language Distribution

- **Translation** → Dominated by English (2380), with smaller counts in Malayalam, Hindi, Bengali, etc.
- **ASR** → Balanced between English (38), Kashmiri (38), Telugu (31).
- **TTS** → Telugu (70) dominates, followed by Maithili (18), English (12).
- **TXT-LANG-DETECTION** → No records.

7. CHALLENGES IDENTIFIED

During the analysis of BHASHADAAN feedback data, several challenges and limitations were identified that may impact both platform evaluation and service improvement efforts.

7.1 Low Alignment Between Automated and User Ratings

One of the most significant findings of the study was the extremely low agreement rate between pipeline-generated ratings and user feedback ratings. Out of 1,632 comparable records, only 20 ratings matched, resulting in a consistency rate of approximately 1.23%.

This indicates that the current automated evaluation framework may not accurately capture user perceptions regarding output quality. Such discrepancies can reduce confidence in system-generated quality metrics and may lead to inaccurate performance assessments.

7.2 Imbalanced Task Distribution

The analysis revealed that Translation accounts for more than 90% of all platform activity, while ASR, TTS, and Language Detection collectively contribute a relatively small percentage of requests.

This imbalance creates challenges when evaluating the effectiveness of less frequently used services, as limited data may not provide statistically reliable insights.

7.3 Limited Usage of Language Detection Services

The Language Detection task recorded only two requests during the observation period. Due to this extremely small sample size, meaningful analysis and performance evaluation could not be conducted.

The low usage may indicate limited user awareness or restricted integration of this functionality within the platform workflow.

7.4 Uneven Language Representation

Although the platform supports multiple Indian languages, user activity remains concentrated around a few dominant languages.

English serves as the primary source language for translation requests, while several regional languages exhibit comparatively low participation. This uneven distribution may affect the development and evaluation of language technologies for underrepresented languages.

7.5 Data Quality and Standardization Challenges

The dataset contained variations in service identifiers, language codes, and naming conventions. Such inconsistencies required preprocessing and standardization before analysis could be performed.

Data quality issues increase analytical complexity and may introduce errors if not properly addressed during preprocessing.

7.6 Limited User Context Information

The available dataset primarily focused on service interactions and ratings. Additional information such as user demographics, geographical distribution, device information, and usage context was not available.

The absence of such information limits the ability to perform deeper behavioral and regional analyses.

8. CONCLUSION

The analysis of BHASHADAAN feedback data provided valuable insights into platform usage patterns, language distribution, service adoption, and quality assessment mechanisms. The findings indicate that Translation services constitute the majority of platform activity, highlighting their importance within the BHASHINI ecosystem. English emerged as the dominant source language, while regional languages demonstrated notable participation in speech-based technologies such as Automatic Speech Recognition (ASR) and Text-to-Speech (TTS).

The study also identified significant differences between pipeline-generated ratings and user feedback ratings. The extremely low agreement rate suggests that the current automated quality evaluation framework may not accurately reflect user satisfaction and requires further refinement. Addressing this gap will be essential for improving service quality monitoring and enhancing user trust in platform outputs.

Additionally, the analysis highlighted opportunities to strengthen regional language participation, promote underutilized services, and improve feedback collection mechanisms. These improvements can contribute to the development of more inclusive, accurate, and user-centric language technologies.

Overall, the BHASHADAAN platform continues to play a crucial role in supporting multilingual AI development in India. Through continuous improvement in quality assessment, user engagement, and language coverage, the platform can further contribute to the vision of enabling accessible and inclusive digital services for speakers of diverse Indian languages.

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8. Internal BHASHADAAN Feedback Dataset used for analysis.
9. Internal platform logs and service metadata collected during the analysis period.
10. Relevant literature on Natural Language Processing (NLP), Machine Translation, Automatic Speech Recognition (ASR), and Text-to-Speech (TTS) systems.